

Prompt 12: Prompt used to obtain the image descriptions in the *Descriptive-iterative* strategy (see Fig. 2c). After the last image, we used the prompt: “That was the last image. Now provide your final answer.”

You'll receive a sequence of images that are a part of a single side of a Bongard Problem. The images will be provided one by one. Your goal is to find a common concept presented in all images. Your description should be as concise as possible. Focus on the most important details. Try to enhance the description of the concept after each image.

The image is always provided correctly. Respond only to the specific request. The first image will be provided in the next message.

Prompt 13: Prompt used for the *Descriptive-iterative* strategy (see Fig. 2c).

You are a vision understanding module designed to provide short, clear and accurate answers. Your goal is to solve the provided Bongard Problem using descriptions of two sides of the problem.

LEFT SIDE DESCRIPTION:
<left_description>

RIGHT SIDE DESCRIPTION:
<right_description>

What is the difference between the two sides of the problem?

Prompt 14: Prompt used to obtain the comparison between the left and right image in the *Contrastive* strategy (see Fig. 2d). After the last image, we used the prompt: “That was the last image. Now provide your final answer.”

You are given two images extracted from the left and right side of a Bongard Problem, respectively. Your goal is to compare the images. Your comparison should be as concise as possible.

Prompt 15: Prompt used for the *Contrastive* and *Contrastive-direct* strategies (see Fig. 2d).

You are a vision understanding module designed to provide short, clear and accurate answers. Your goal is to solve the provided Bongard Problem using comparisons between pairs of images. Each pair contains one image from the left and one from the right side of the problem.

COMPARISONS:
<comparisons>

What is the difference between the two sides of the problem?

Prompt 16: Prompt used for the *Contrastive-iterative* strategy (see Fig. 2e). After the last pair of images, we used the prompt: “It was the last pair of images. What is the difference between the two sides of the problem?”

You are a vision understanding module designed to provide short, clear and accurate answers. Your goal is to solve the provided Bongard Problem. You'll receive a sequence of image pairs. Each pair contains one image from the left and one from the right side of the problem. In each step compare the two images and refine the definitions of concepts that describe left and right sides of the problem. Your description should be as concise as possible. Focus on the most important details. The first pair will be provided in the next message.

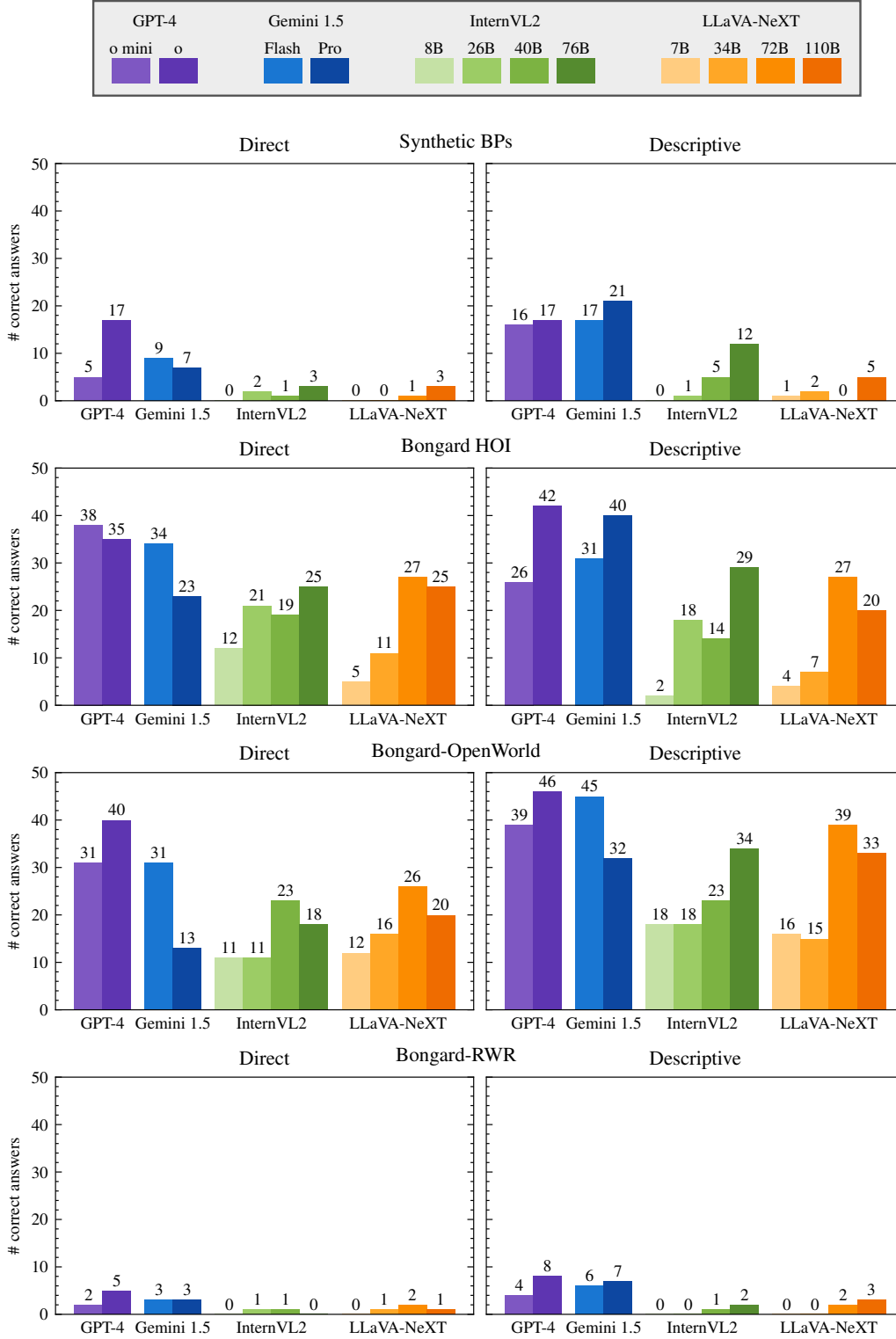


Figure 10: **The impact of model scaling on abstract reasoning abilities.** We considered a diverse set of model sizes across proprietary and open-access MLLMs. The experiments cover all 4 datasets using the Direct and Descriptive solution strategies.

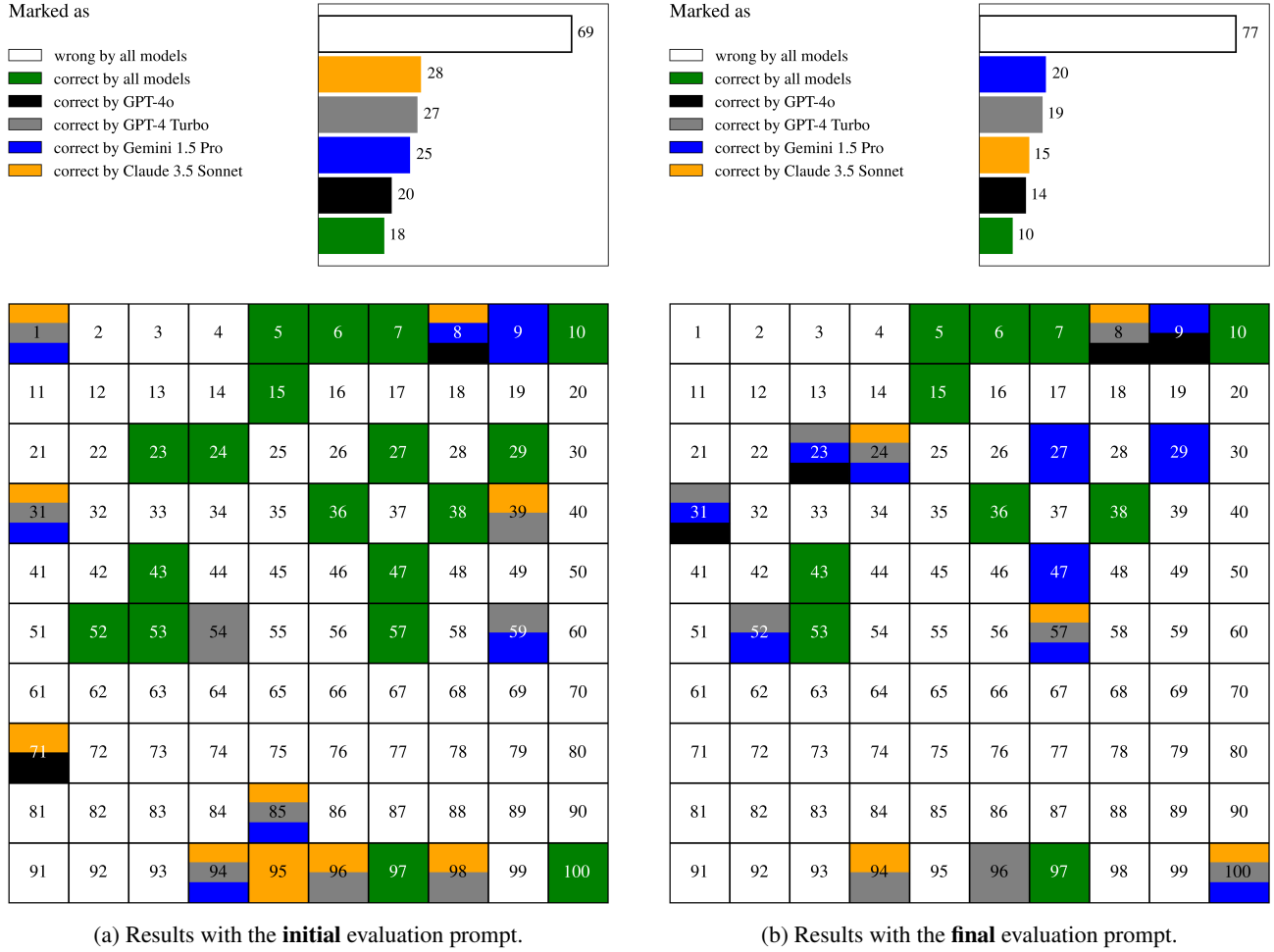


Figure 11: **Models’ agreement on the evaluation of BPs.** The assessed solutions were generated by GPT-4o with the *Descriptive* strategy. The numbers refer to the BP tasks from (Bongard, 1968). Green indicates tasks unanimously evaluated as correctly solved by all models, while white indicates unanimous incorrect evaluations. Other colors highlight tasks marked as correctly solved by individual models.